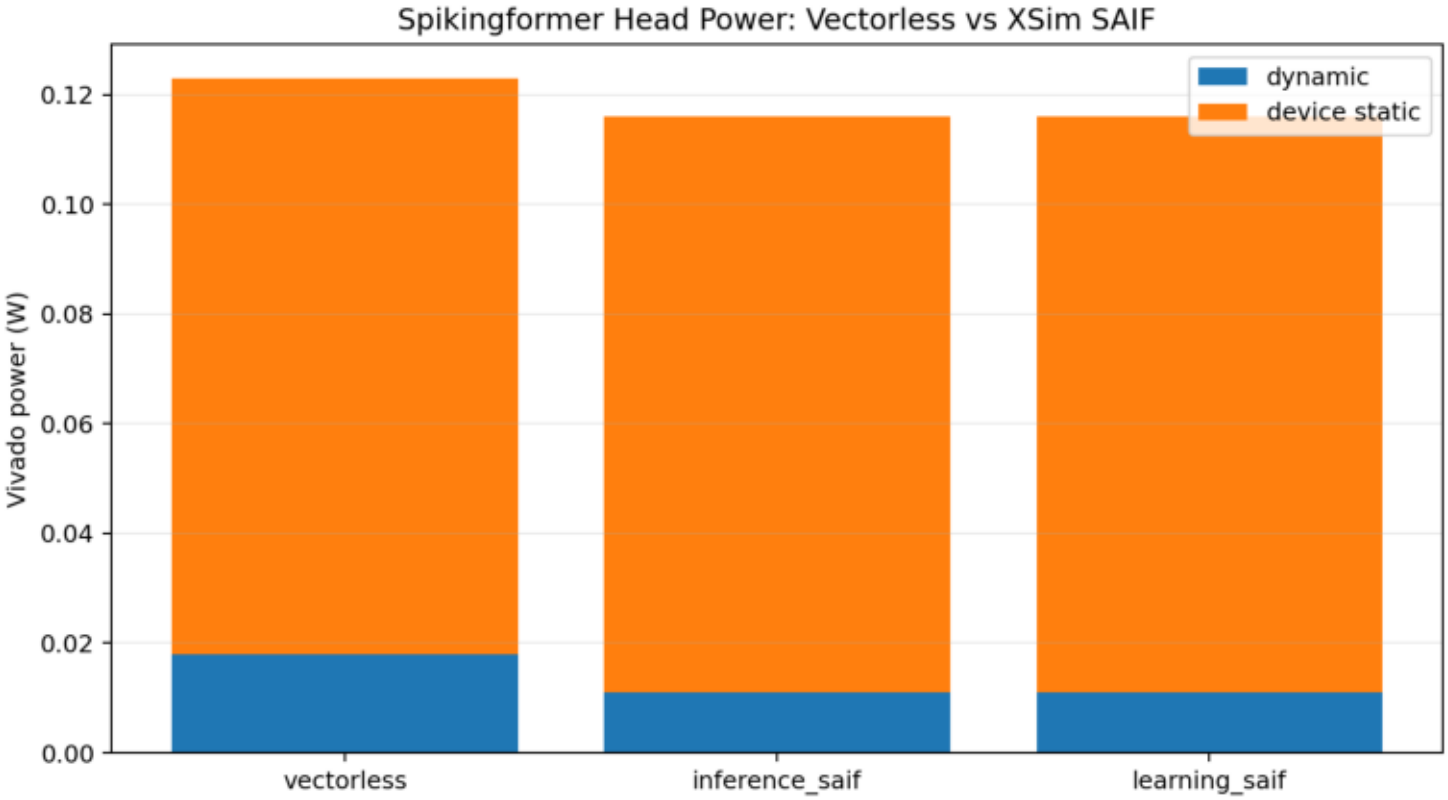

Spikingformer Head Power: Vectorless vs XSim SAIF
dynamic
device static
Vivado power (W)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
vectorless
inference_saif
learning_saif

Spikingformer Head Activity Power Summary


```
{
  "note": "Vivado-estimated FPGA power for the routed Spikingformer-head accelerator. Vectorless is Vivado's default activity estimate; inference_saif and learning_saif are annotated from XSim SAIF traces of the RTL testbenches.",
  "target": {
    "part": "xc7z020clg400-1",
    "clock_ns": 10.0,
    "top": "cifar10_spikingformer_head_pynq_axi_stream_top",
    "annotated_instance": "u_head"
  },
  "validation": {
    "samples": 10000,
    "accuracy": 0.9551,
    "score_mismatch_count": 0,
    "prediction_mismatch_count": 0,
    "onchip_learning_updates": 25,
    "onchip_learning_score_mismatch_count": 0,
    "onchip_learning_prediction_mismatch_count": 0
  },
  "vectorless": {
    "report": "C:\\Users\\Ross\\Downloads\\neuraedge--main\\neuraedge--main\\cifar10_neuromorphic\\outputs_spikingformer_head_accelerator\\vivado\\spikingformer_head_power.rpt",
    "total_on_chip_w": 0.123,
    "dynamic_w": 0.018,
    "device_static_w": 0.105,
    "junction_temp_c": 26.4,
    "confidence": "Low"
  },
  "inference_saif": {
    "report": "C:\\Users\\Ross\\Downloads\\neuraedge--main\\neuraedge--main\\cifar10_neuromorphic\\outputs_spikingformer_head_accelerator\\activity_power\\spikingformer_head_power_inference_saif.rpt",
    "total_on_chip_w": 0.116,
    "dynamic_w": 0.011,
    "device_static_w": 0.105,
    "junction_temp_c": 26.3,
    "confidence": "Medium"
  },
  "learning_saif": {
    "report": "C:\\Users\\Ross\\Downloads\\neuraedge--main\\neuraedge--main\\cifar10_neuromorphic\\outputs_spikingformer_head_accelerator\\activity_power\\spikingformer_head_power_learning_saif.rpt",
    "total_on_chip_w": 0.116,
    "dynamic_w": 0.011,
    "device_static_w": 0.105,
    "junction_temp_c": 26.3,
    "confidence": "Medium"
  },
  "inference_saif_dynamic_vs_vectorless": 0.6111111111111112,
  "inference_saif_total_vs_vectorless": 0.943089430894309,
  "learning_saif_dynamic_vs_vectorless": 0.6111111111111112,
  "learning_saif_total_vs_vectorless": 0.943089430894309
}
```